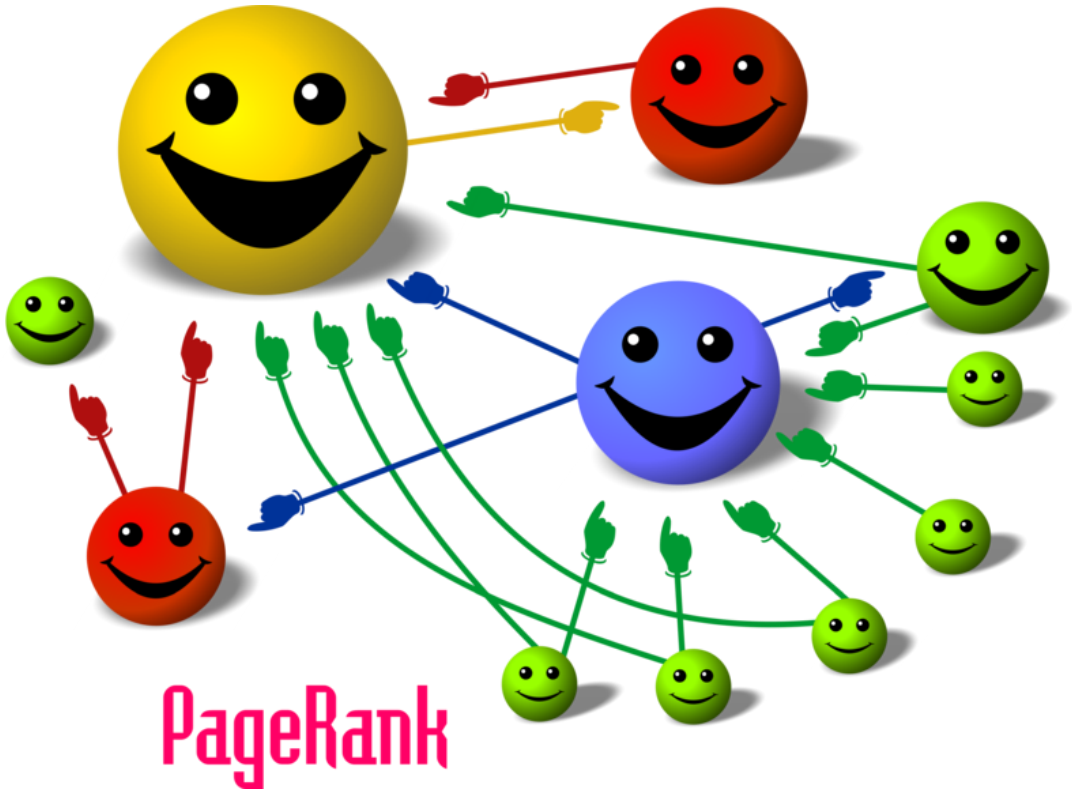
Link Analysis for Web Search



Slides by Raed Almomani and Andrew DeOrio

Review

- Last time: ranked pages using page content (words)
  - Based on what appears in the document
- Today: improve page rank using links between web pages
  - Based on document importnace

Agenda

- **Document importance**
- Link-based ranking algorithms
  - PageRank algorithm
  - HITS algorithm
- Search engine optimization

Importance in the real world

- How do people indicate something is important?
  - Talk about it
  - Cite it (academic papers)
  - Upvote it (Reddit)
  - Link to it

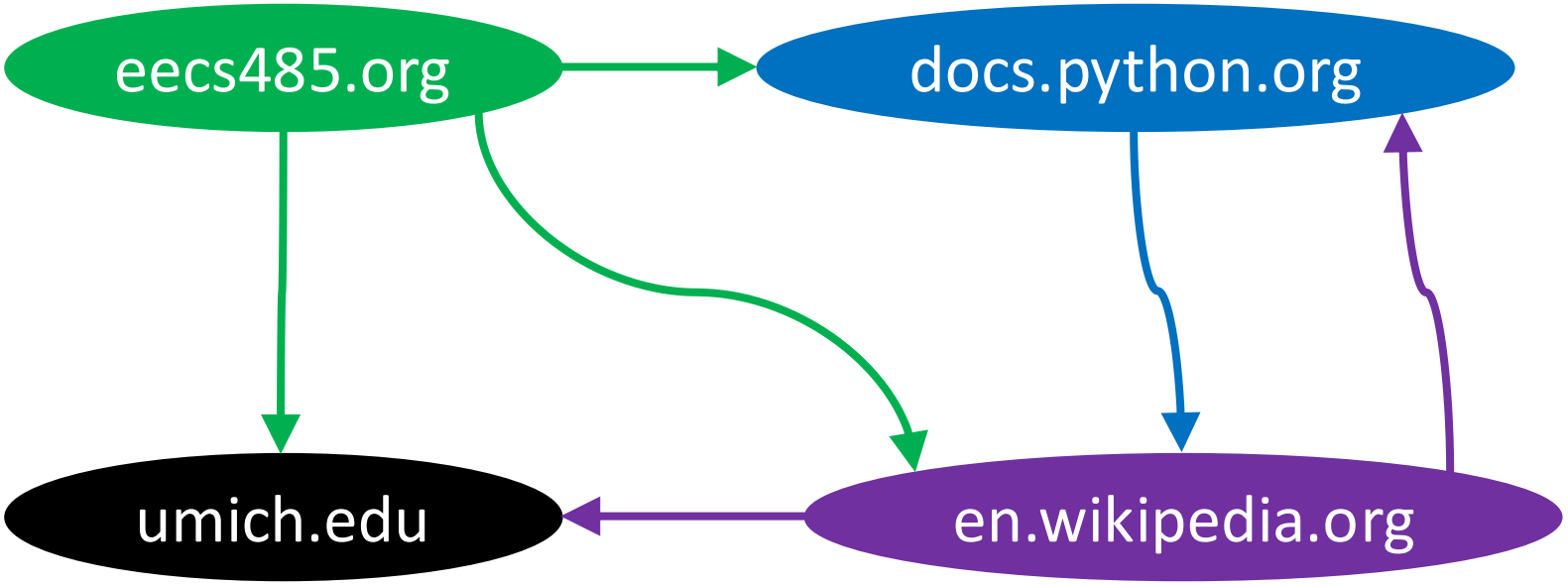
Humans indicate importance

- Humans know better than computers which web pages are important

- On the web, humans indicate importance using **links**
  - Similar to **citations in academic papers**

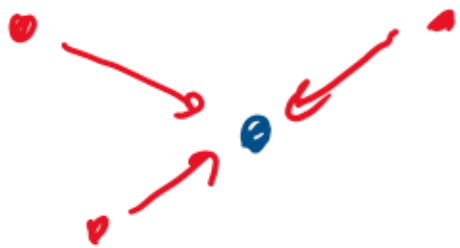
Link graph

- Web as a *graph* (AKA *network*)
  - Each web page is a node
  - Each hyperlink is a directed edge



Using the link graph for importance

- If many pages link to a page, it is likely important



- If an important page links to a page, that page also is important

Web search history

- Search in the 1990s was pretty bad
  - Hand-curated list (e.g. Yahoo)
  - Search engines used only page content (e.g. AltaVista)
- Late 1990s: improve search using hyperlink graph
  - **PageRank**, Page & Brin
  - **HITS**, Kleinberg

More on the Wikipedia Timeline of web search engines

Page importance and search results

Goal:

1. Make the web as graph
2. Compute important scores for pages
3. Use those scores to rank search results higher

Agenda

- Document importance
- Link-based ranking algorithms
  - **PageRank algorithm**
  - HITS algorithm
- Search engine optimization

PageRank

- Algorithm that made Google famous

The PageRank Citation Ranking: Bringing Order to the Web

January 29, 1998

Abstract

The importance of a Web page is an inherently subjective matter, which depends on the readers interests, knowledge and attitudes. But there is still much that can be said objectively about the relative importance of Web pages. This paper describes PageRank, a method for rating Web pages objectively and mechanically, effectively measuring the human interest and attention devoted to them.

We compare PageRank to an idealized random Web surfer. We show how to efficiently compute PageRank for large numbers of pages. And, we show how to apply PageRank to search and to user navigation.

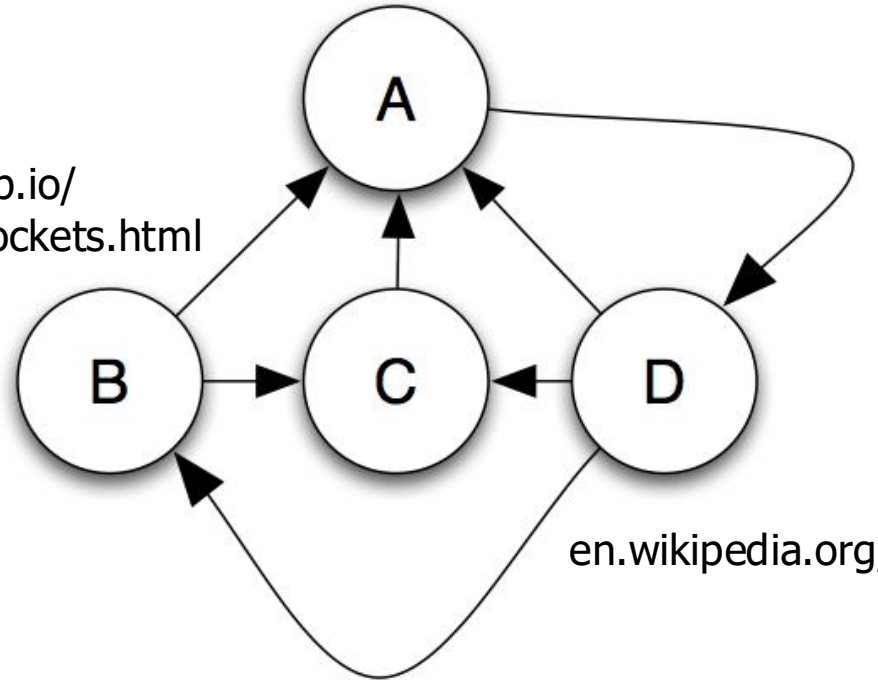
PageRank model

- Assume someone starts on a random web page
- Click a random link on that page
- Over and over
- Web pages with more visits are more important

Random surfer example

python.org/3/library/threading.html

eecs485staff.github.io/
p4-mapreduce/threads-sockets.html



en.wikipedia.org/wiki/Thread_(computing)

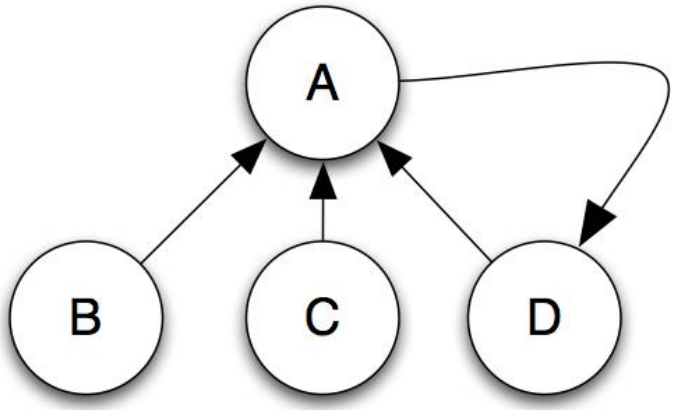
pythoncentral.io/how-to-create-a-thread-in-python/

PageRank

- A node with C outgoing links contributes 1/C of its PageRank to each linked node

$$PR(A) = \frac{(1-d)}{N} + d \sum_i \frac{PR(I_i)}{C(I_i)}$$

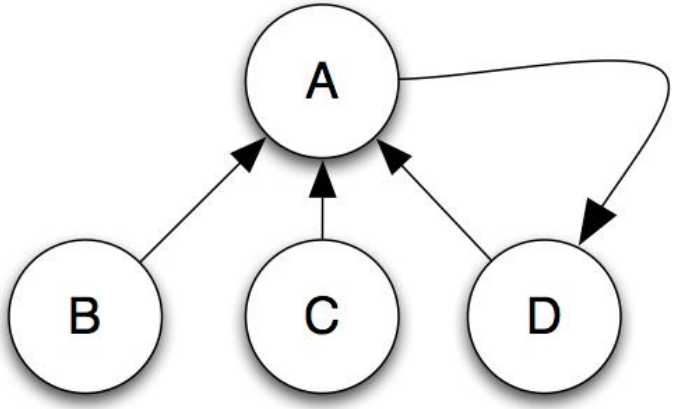
- Damping factor d is usually 0.85



$$PR(A) = \frac{(1-d)}{N} + d \sum_i \frac{PR(I_i)}{C(I_i)}$$

PageRank example

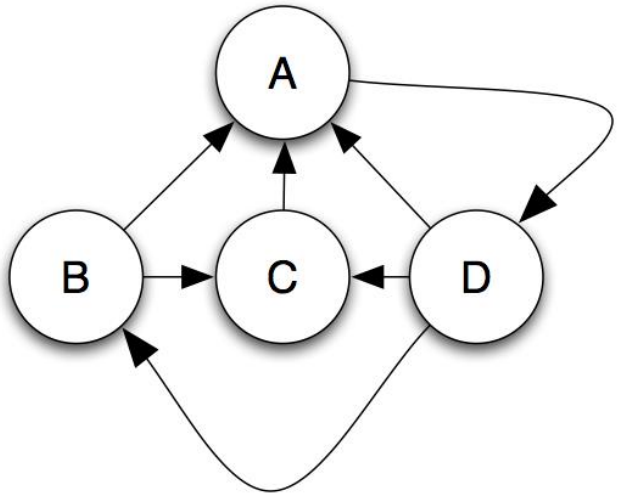
- Total PR = 1, so initialize each node to 0.25
- Set d = 0.85
- PR(A) = (1 - 0.85)/4 +
  0.85 * (0.25/1 + 0.25/1 + 0.25/1)
- PR(A) = 0.675



$$PR(A) = \frac{(1-d)}{N} + d \sum_i \frac{PR(I_i)}{C(I_i)}$$

PageRank example

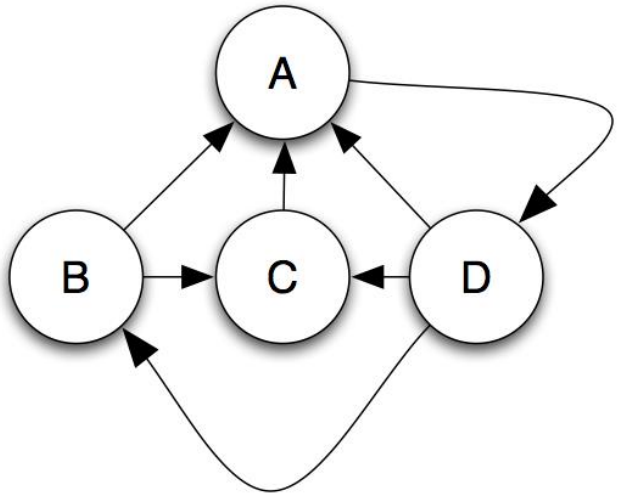
- Again, initialize all nodes to 0.25 and d=0.85
- PR(A) = *compute this*



$$PR(A) = \frac{(1-d)}{N} + d \sum_i \frac{PR(I_i)}{C(I_i)}$$

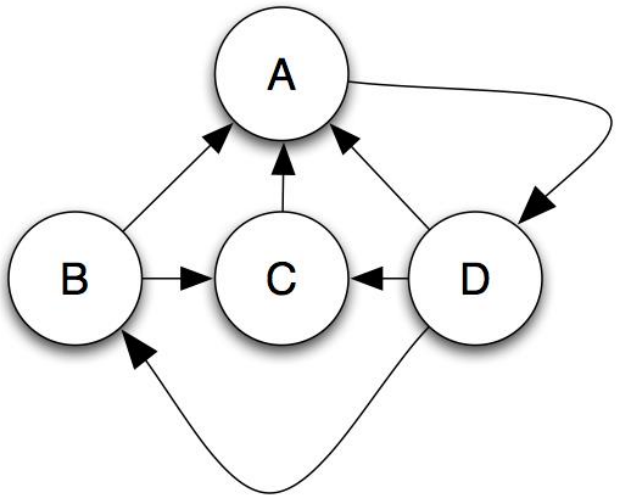
PageRank example

- Again, initialize all nodes to 0.25 and d=0.85
- PR(A) = (1 - 0.85)/4 +
  0.85 * (0.25/2 + 0.25/1 + 0.25/3)
  = .0375+ .85*(0.125 + 0.25 + 0.083)
  = 0.4268



Example

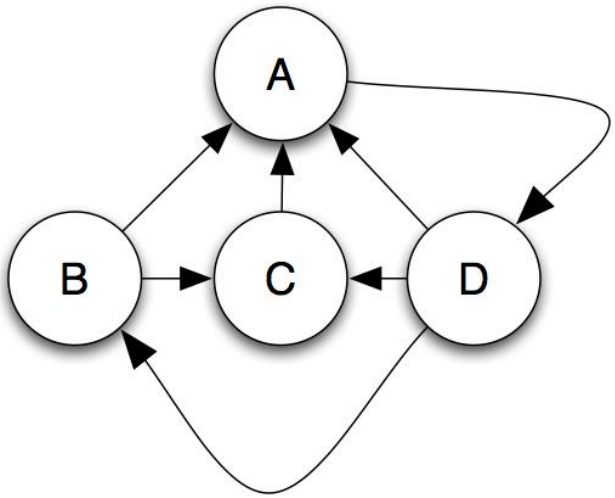
$$PR(A) = \frac{(1-d)}{N} + d \sum_i \frac{PR(I_i)}{C(I_i)}$$



A	**B**	**C**	**D**
0.25	0.25	0.25	0.25

Example

$$PR(A) = \frac{(1-d)}{N} + d \sum_i \frac{PR(I_i)}{C(I_i)}$$

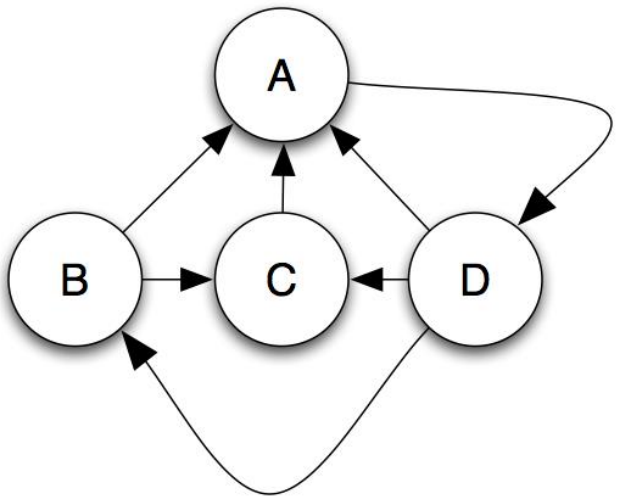


PR(A) = 0.0375 + 0.85(0.25/2 + 0.25/1 + 0.25/3)

A	**B**	**C**	**D**
0.25	0.25	0.25	0.25
0.428			

Example

$$PR(A) = \frac{(1-d)}{N} + d \sum_i \frac{PR(I_i)}{C(I_i)}$$

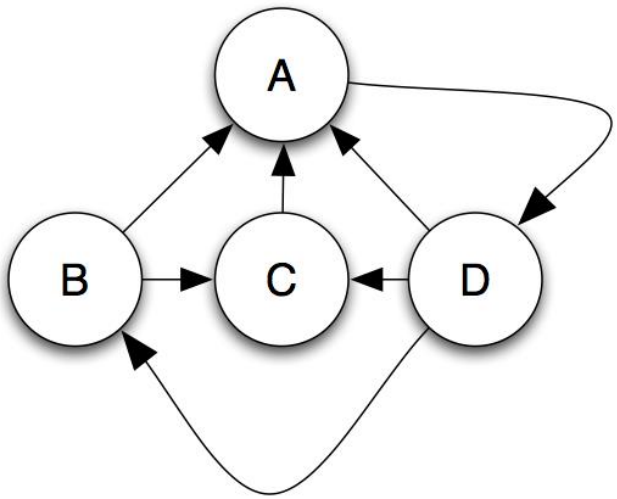


PR(B) = 0.0375 + 0.85(0.25/3)

A	**B**	**C**	**D**
0.25	0.25	0.25	0.25
0.428	0.109		

Example

$$PR(A) = \frac{(1-d)}{N} + d \sum_i \frac{PR(I_i)}{C(I_i)}$$

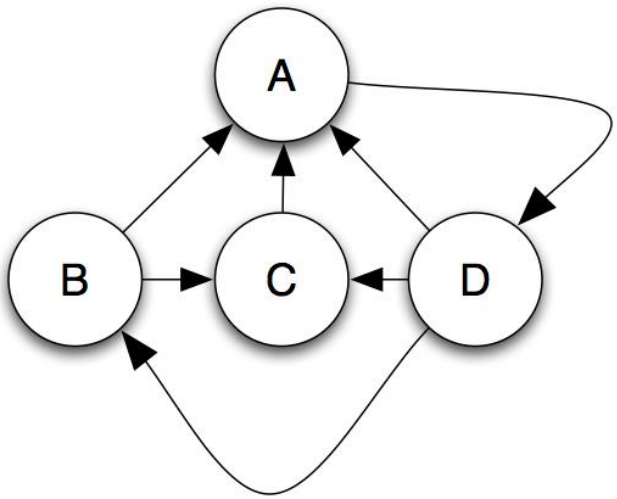


PR(C) = 0.0375 + 0.85(0.25/2 + 0.25/3)

A	**B**	**C**	**D**
0.25	0.25	0.25	0.25
0.428	0.109	0.215	

Example (normalization)

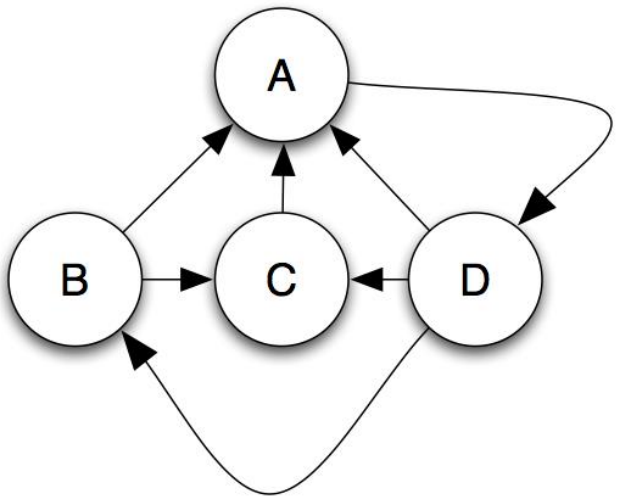
$$PR(A) = \frac{(1-d)}{N} + d \sum_i \frac{PR(I_i)}{C(I_i)}$$



A	B	C	D
0.25	0.25	0.25	0.25
0.428	**0.109**	**0.215**	**0.25**

Example

$$PR(A) = \frac{(1-d)}{N} + d \sum_i \frac{PR(I_i)}{C(I_i)}$$

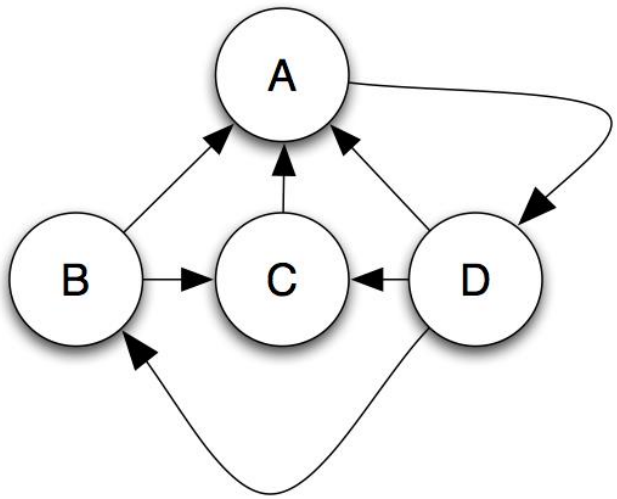


PR(D) = 0.0375 + 0.85(0.25/1)

A	**B**	**C**	**D**
0.25	0.25	0.25	0.25
0.427	0.108	0.215	0.25

Example

$$PR(A) = \frac{(1-d)}{N} + d \sum_i \frac{PR(I_i)}{C(I_i)}$$

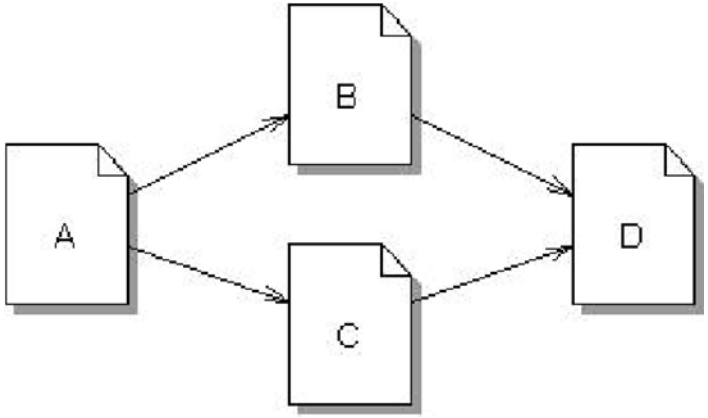


A	**B**	**C**	**D**
0.25	0.25	0.25	0.25
0.427	0.108	0.215	0.25
0.337	0.108	0.154	0.401
0.328	0.151	0.197	0.324
0.361	0.129	0.193	0.317

$$PR(A) = \frac{(1-d)}{N} + d \sum_i \frac{PR(I_i)}{C(I_i)}$$

Sink regions

- What happens after many iterations?
- Ignore damping factor (1-d)/N
- PageRank “drains” into sink node (page with no outgoing links)



Damping factor to solve sink region

- Must have non-zero probability of reaching every node from every other node
- Solution: Add prob(1-d)/N of jumping to any page

$$PR(A) = \frac{(1-d)}{N} + d \sum_{i} \frac{PR(I_i)}{C(I_i)}$$

Damping factor intuition

- Probability of the user randomly jump to any page in the network rather than following a link.

Adding PageRank to a search engine

- Weighted sum of page importance and query-similarity
- Score(query, doc)=
  - w*sim(q, p) + (1-w) * PR(p)
    - If sim(q, p) > 0
    - Otherwise, 0
- Where:
  - 0 < w < 1
  - Values sim(q,p) and PR(p) are normalized

Agenda

- Document importance
- Link-based ranking algorithms
  - PageRank algorithm
  - **HITS algorithm**
- Search engine optimization

Hubs and authorities

- Unlike PageRank, HITS is query-dependent

- A page is a good ***authority*** if it is pointed-to by many good ***hubs***
- A page is a good ***hub*** if it points to many good ***authorities***
- Good hubs and authorities reinforce each other

HITS algorithm

$$auth(p) = \sum_{i=1}^{n} hub(i)$$

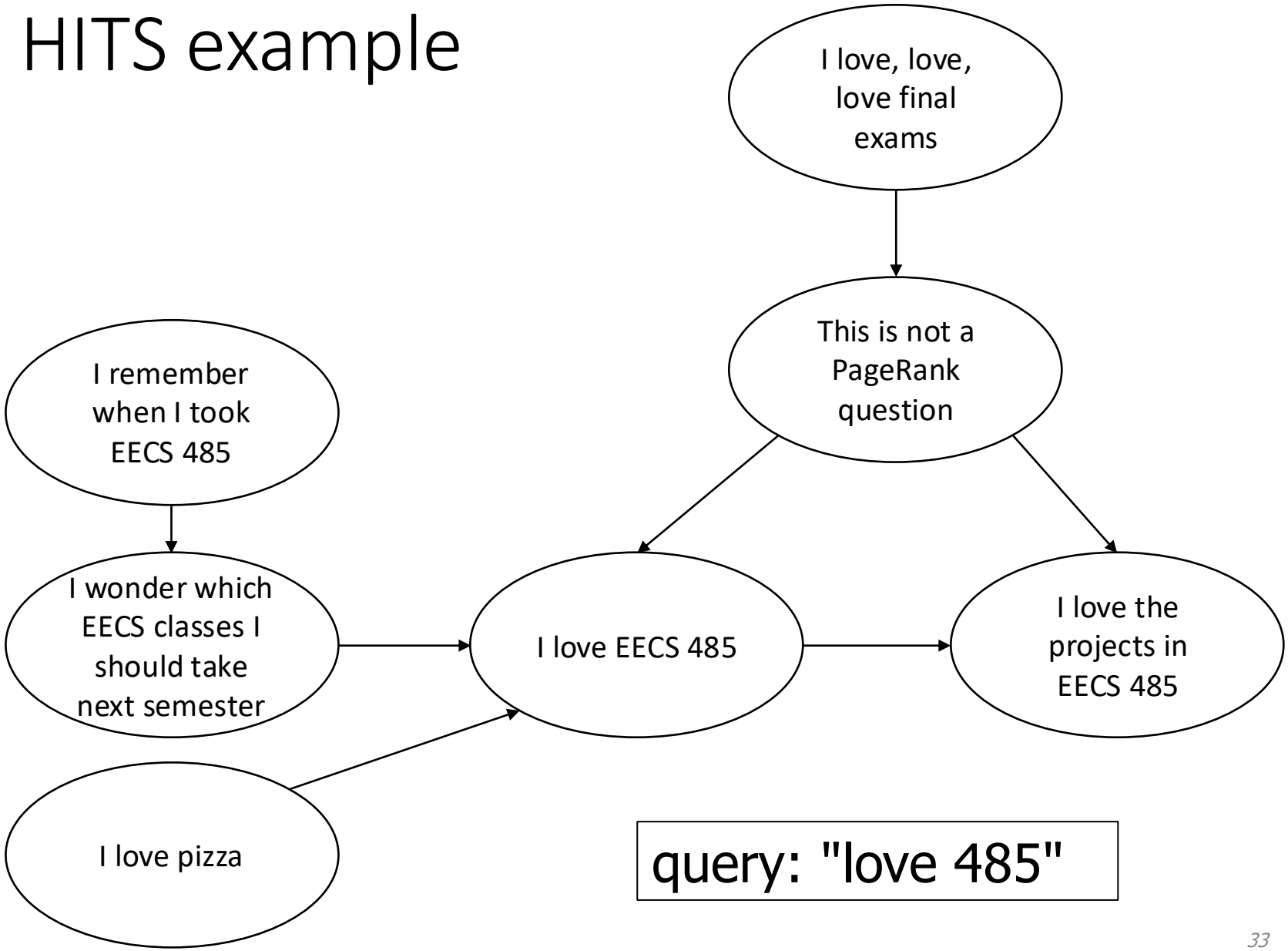
$$hub(p) = \sum_{i=1}^{n} auth(i)$$

1. Obtain *root set* using input query
2. Expand the root set by radius 1. This is called the *base set*
3. Iteratively compute hub and authority scores for each node in graph

More HITS

1. Initialize all auth and hub scores to 1
2. For all nodes, update authority scores
3. For all nodes, update hub scores
4. Normalize scores
   - Divide each auth by sqrt(sum($auth^2$))
   - Divide each hub by sqrt(sum(hub^2))
5. If converges, terminate; else goto 2

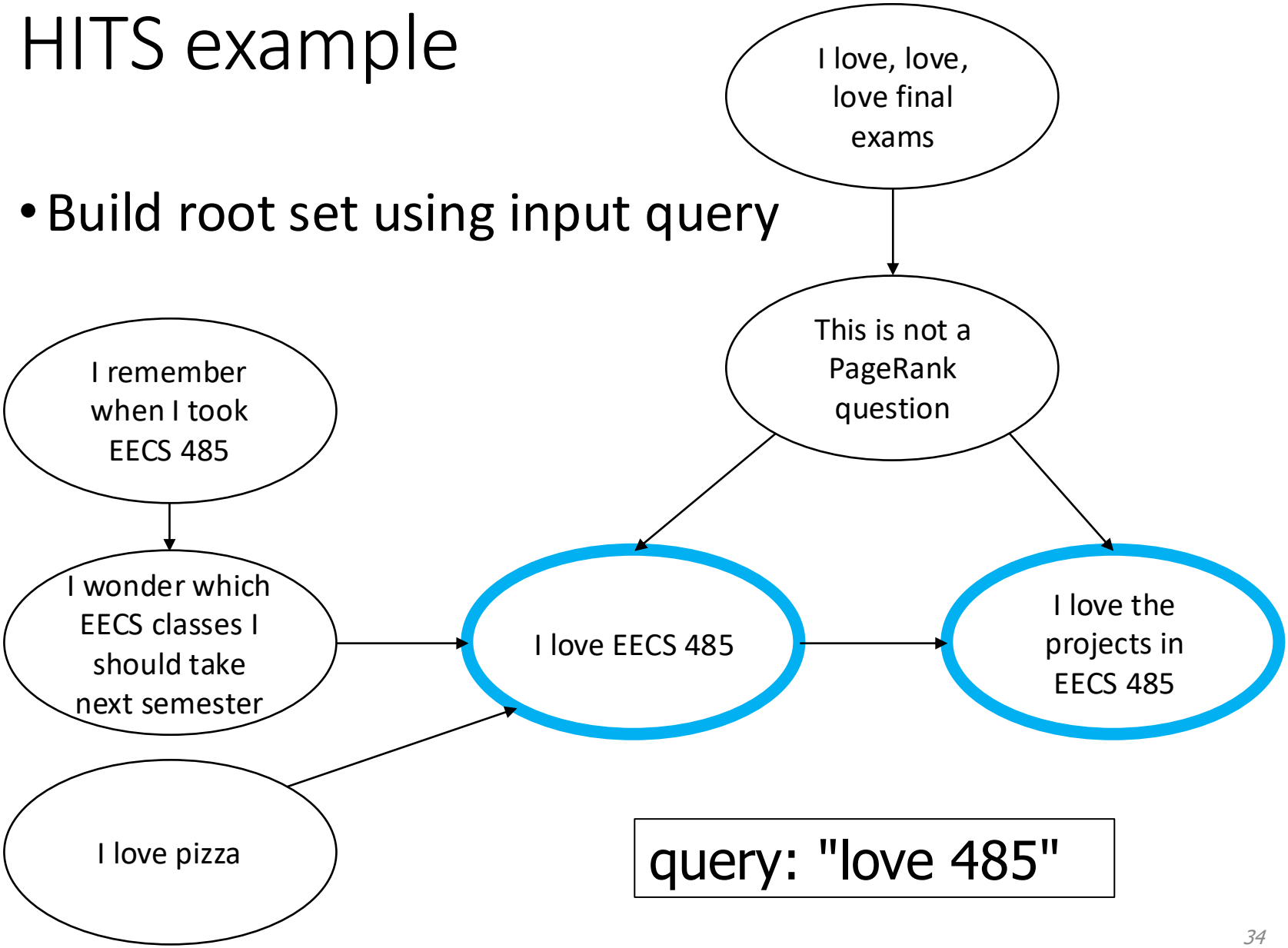
HITS example



query: "love 485"

HITS example

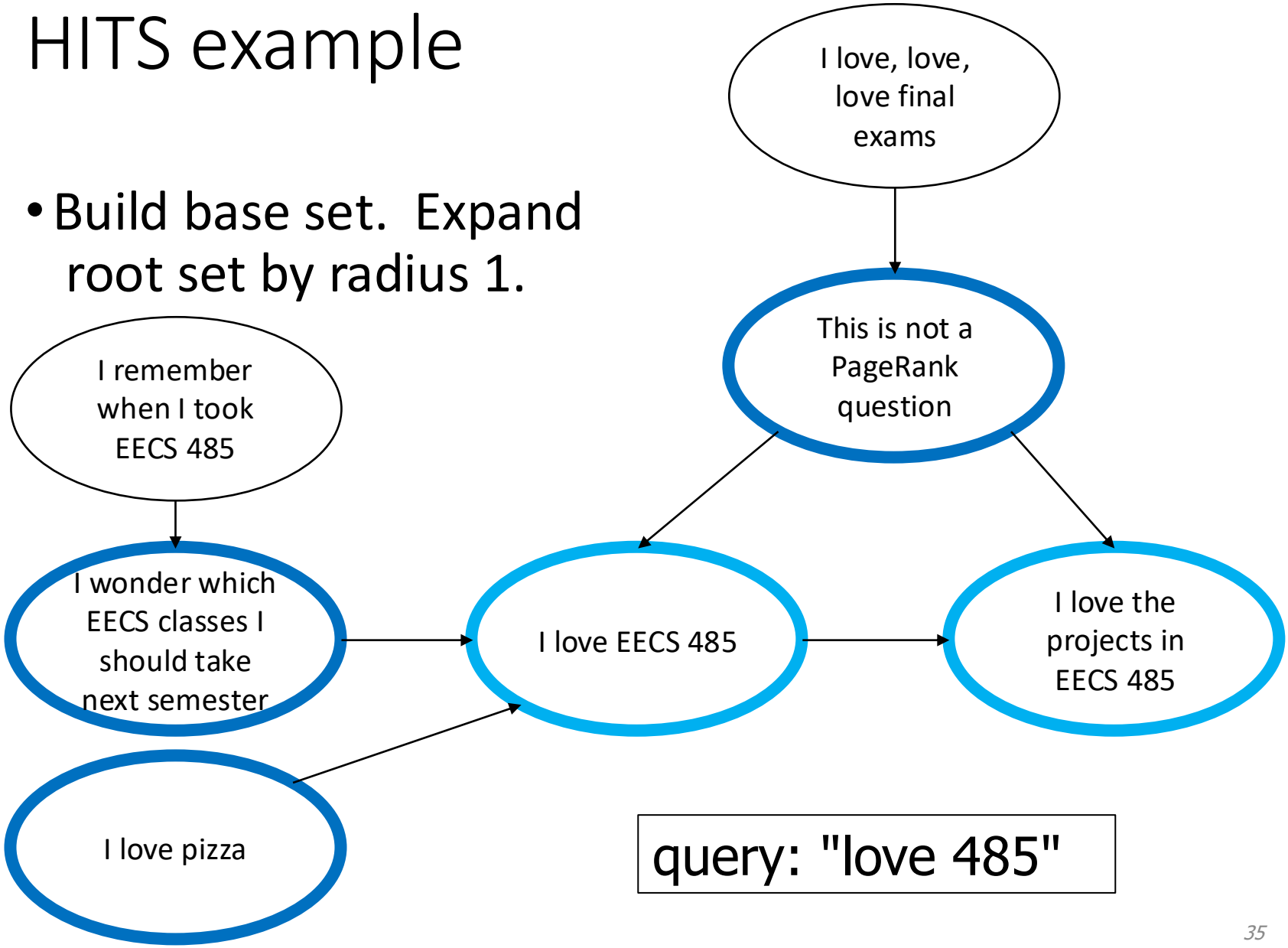
- Build root set using input query



query: "love 485"

HITS example

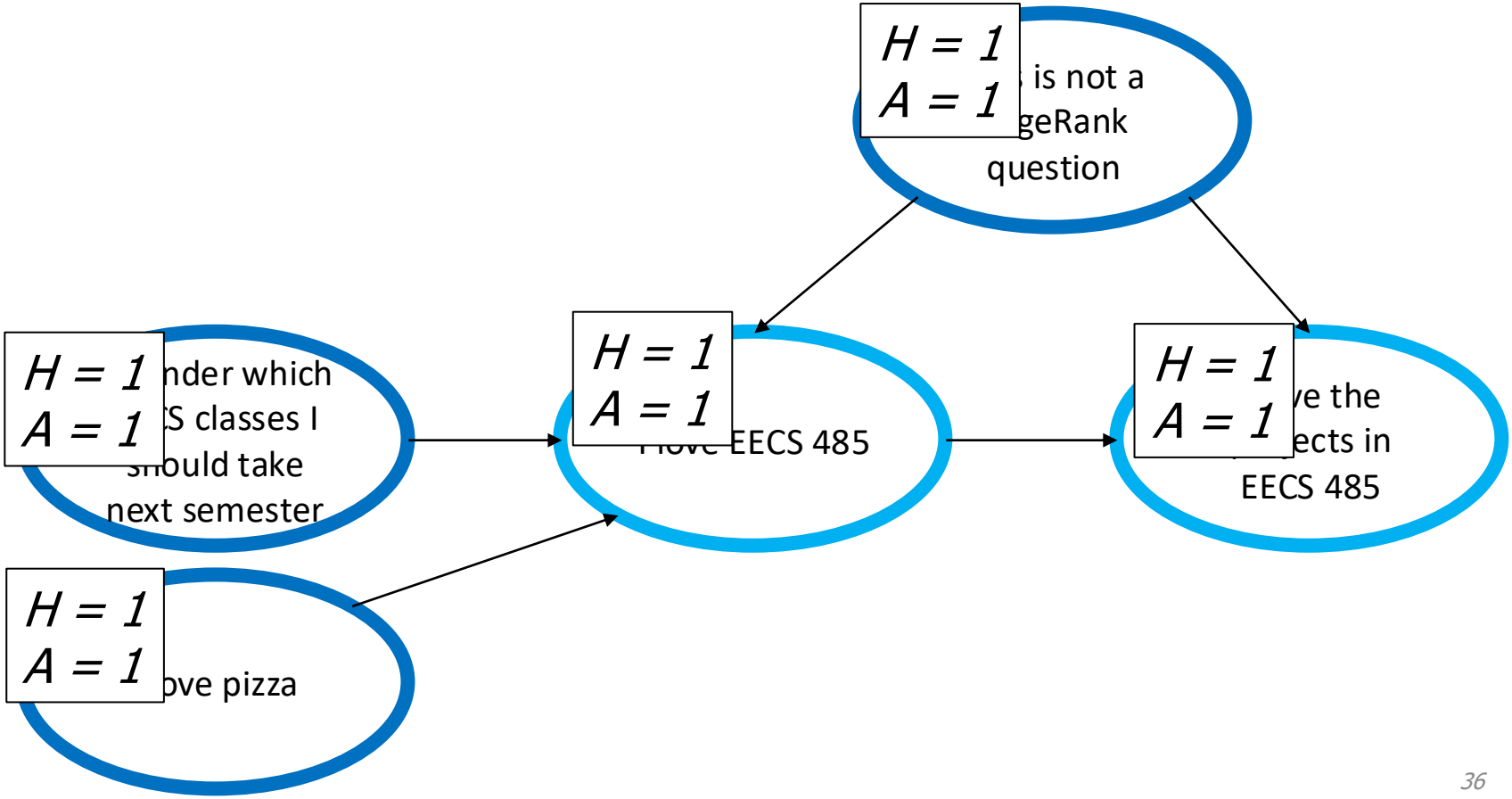
- Build base set. Expand root set by radius 1.



query: "love 485"

HITS example

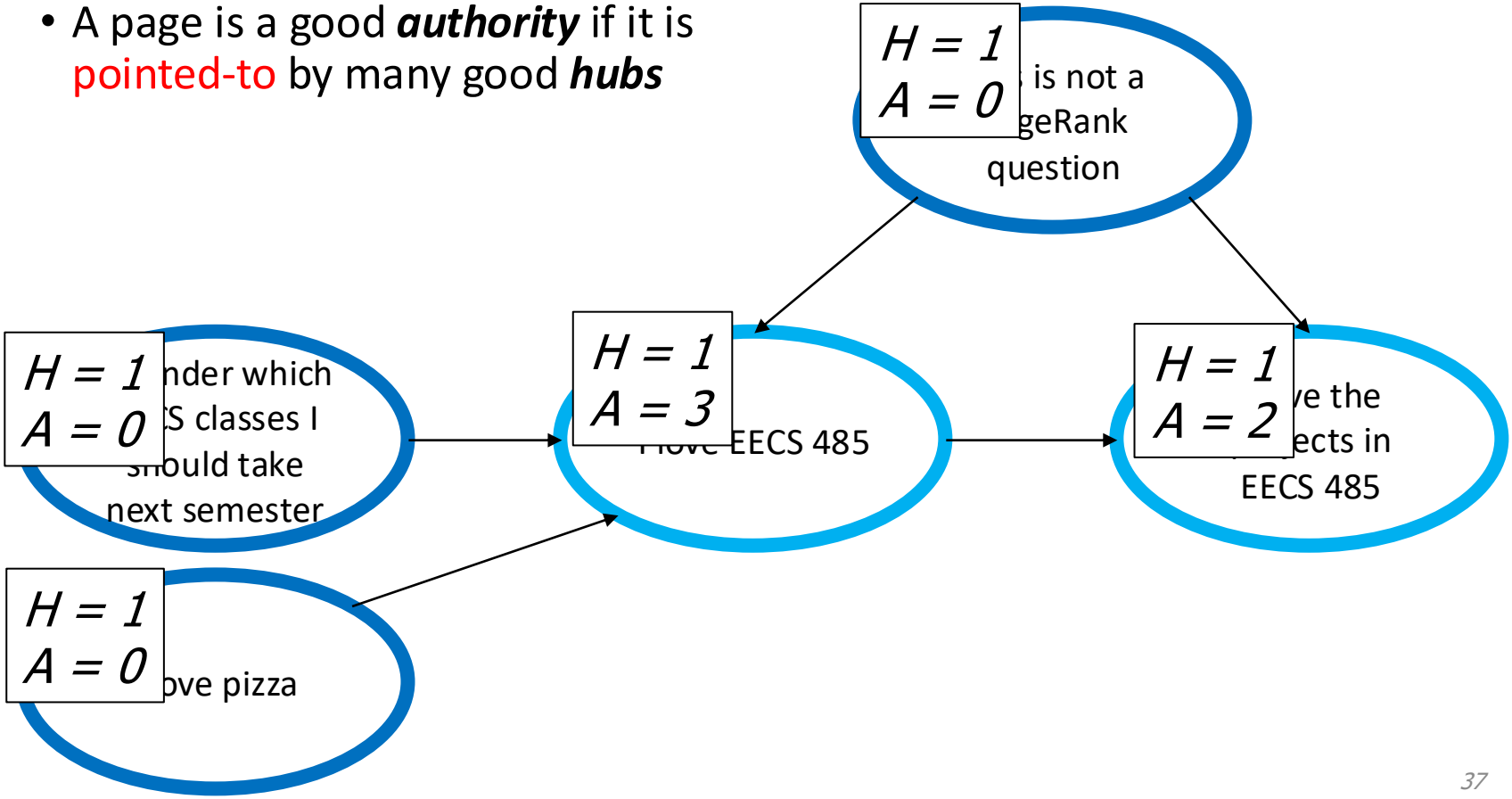
- Initialize all hub and authority scores to 1



$$auth(p) = \sum_{i=1}^{n} hub(i)$$

HITS example

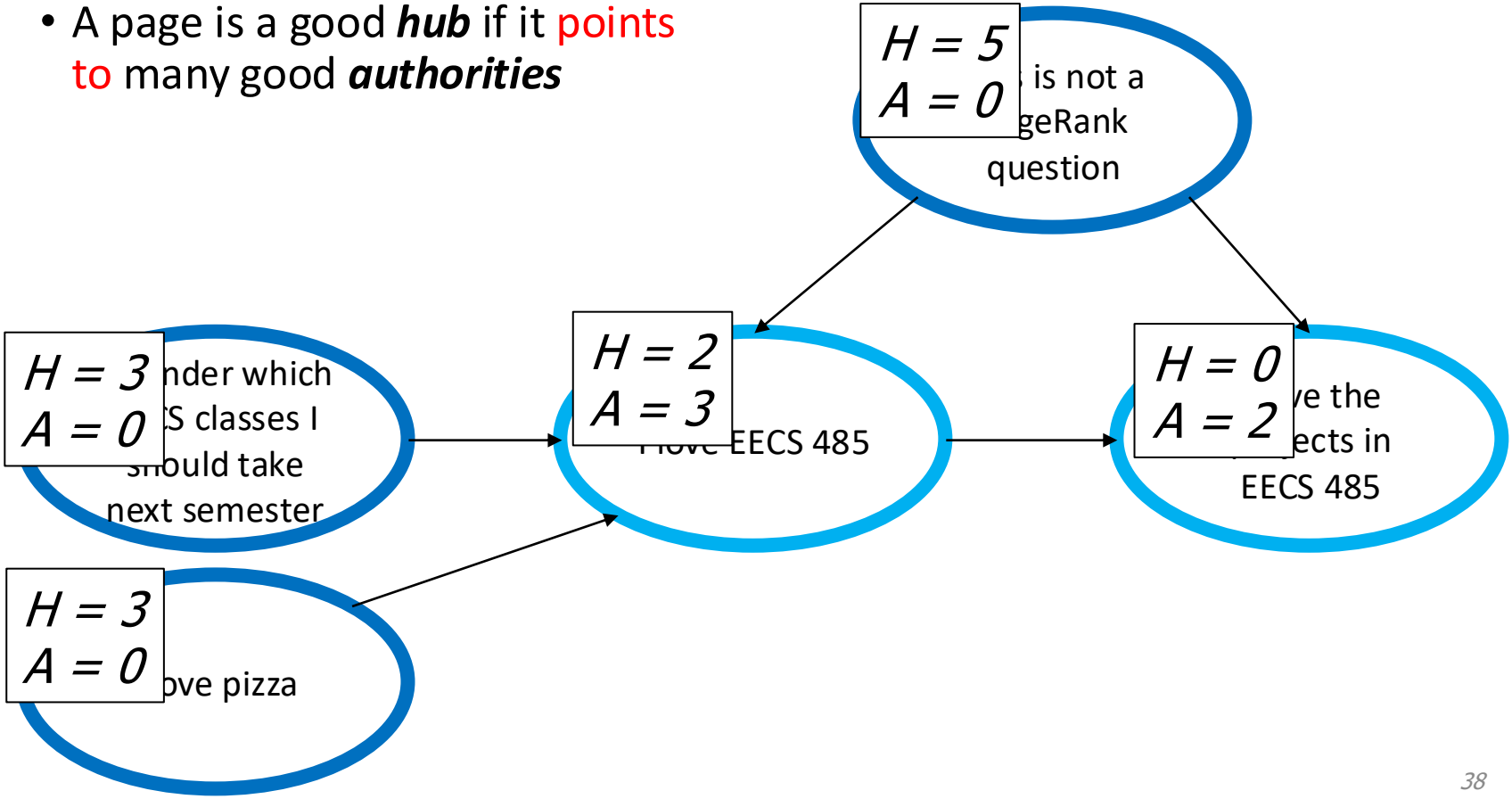
- Update authority scores
  - A page is a good ***authority*** if it is pointed-to by many good ***hubs***



$$hub(p) = \sum_{i=1}^{n} auth(i)$$

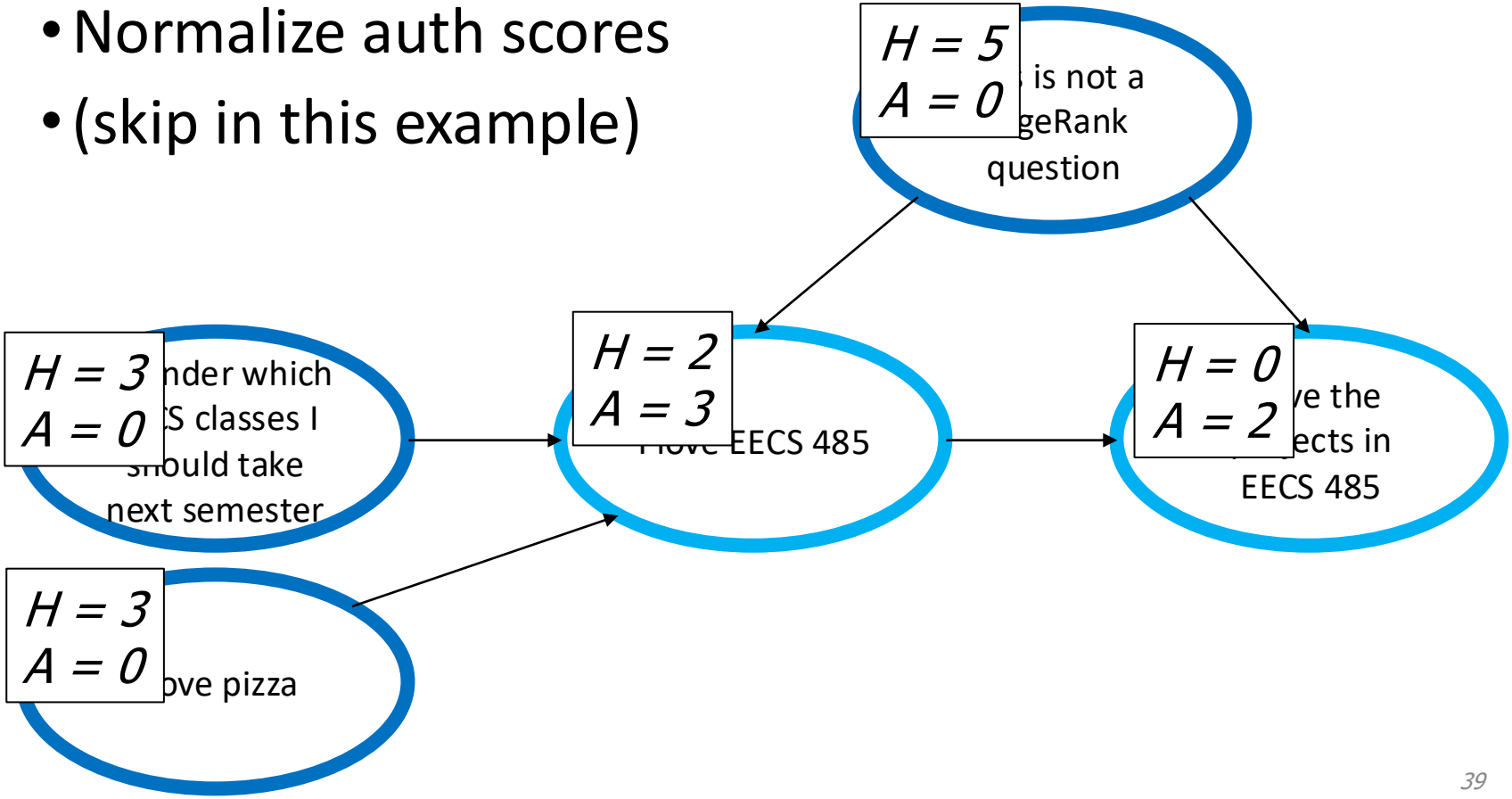
HITS example

- Update hub scores
  - A page is a good ***hub*** if it points to many good ***authorities***



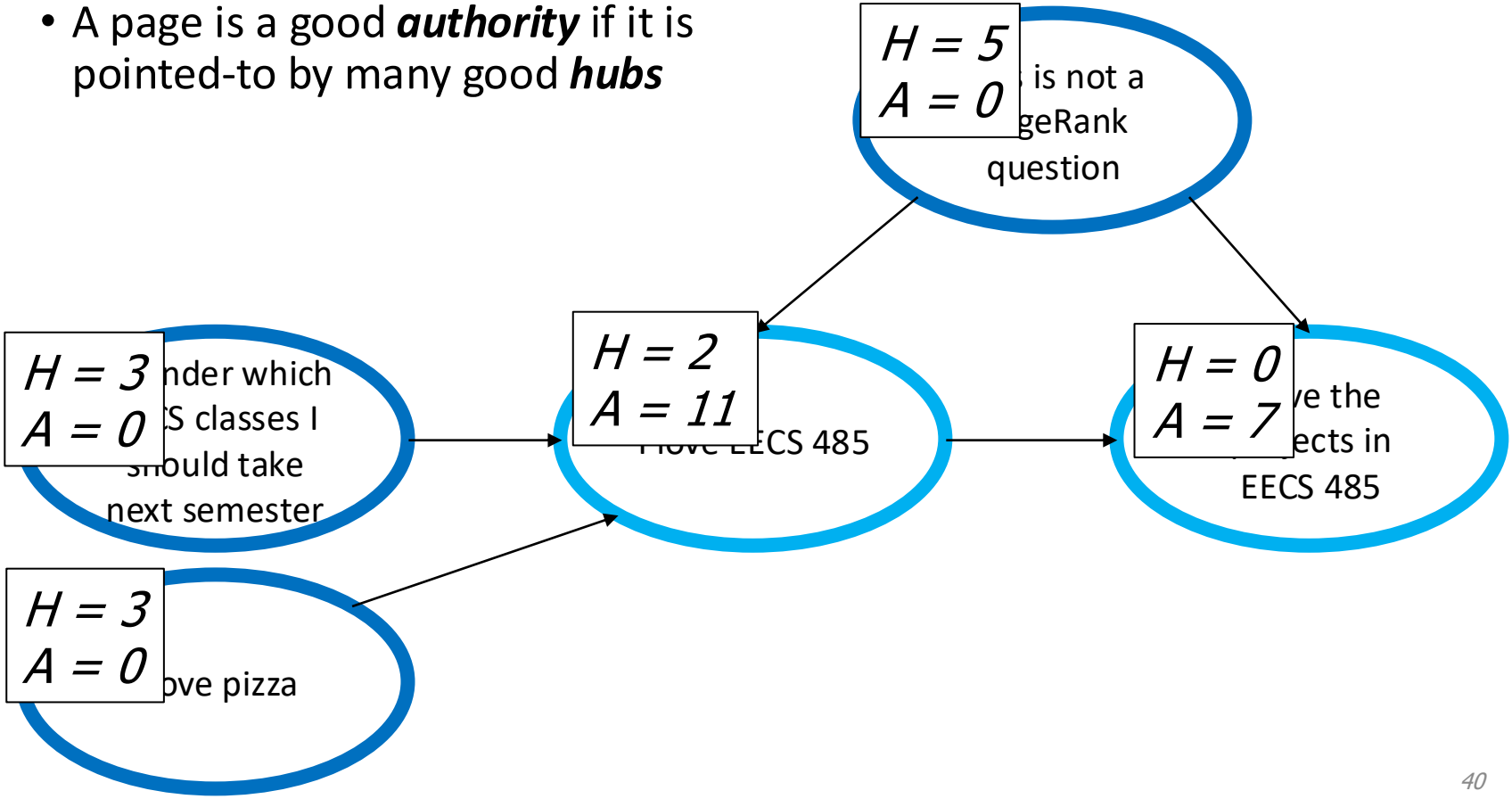
HITS example

- Normalize hub scores
- Normalize auth scores
- (skip in this example)



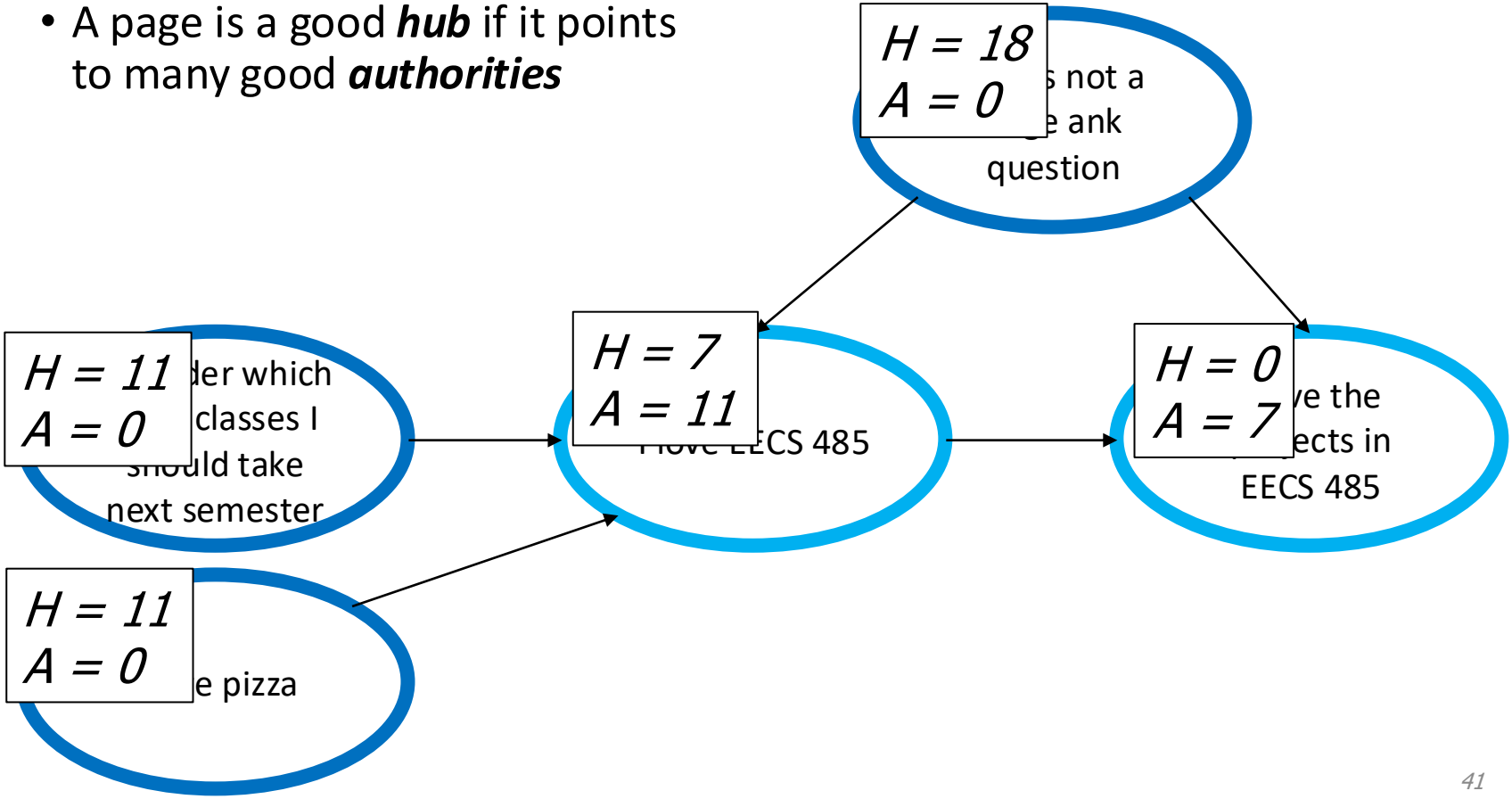
HITS example

- Update authority scores
  - A page is a good ***authority*** if it is pointed-to by many good ***hubs***



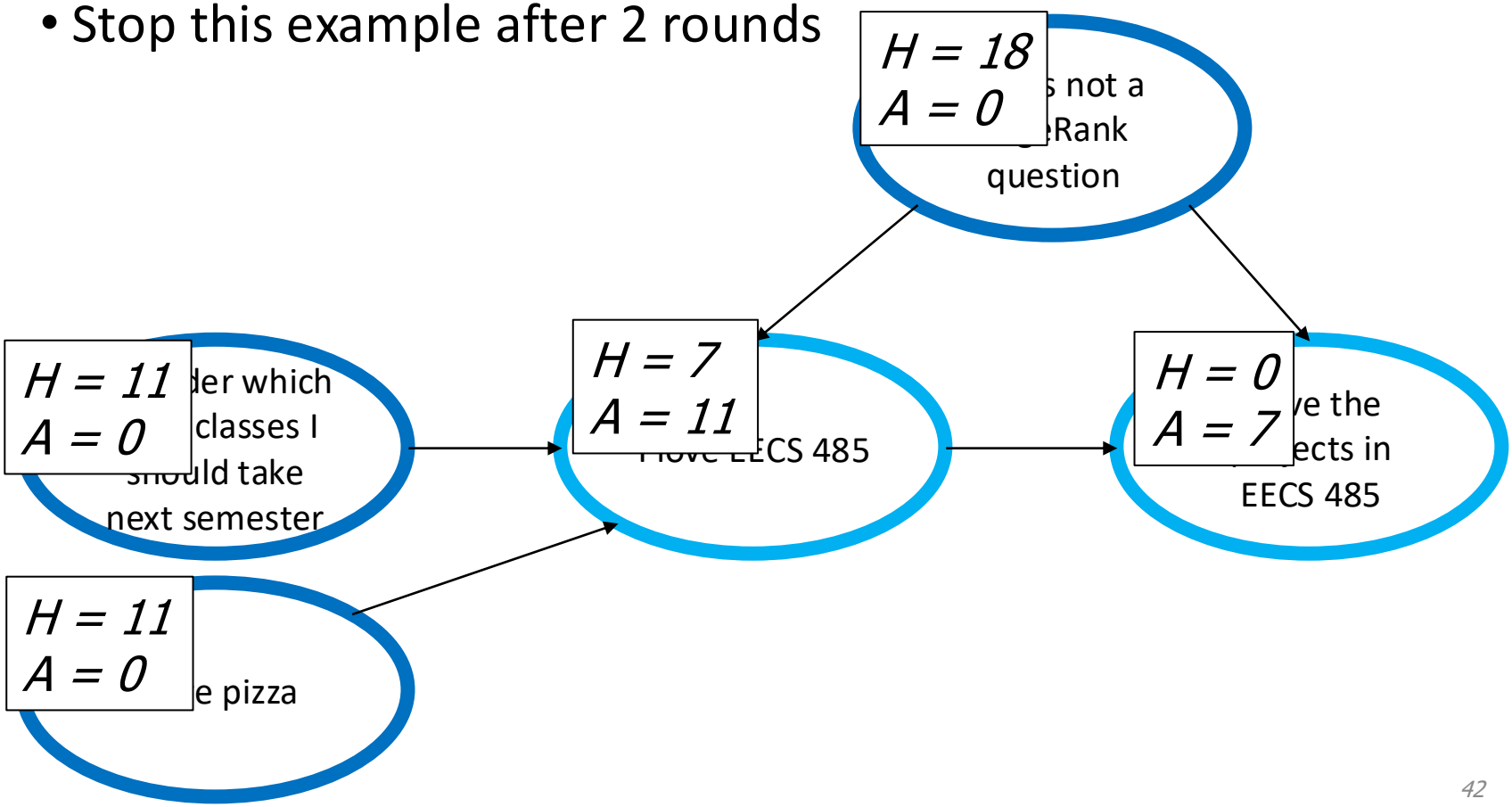
HITS example

- Update hub scores
  - A page is a good ***hub*** if it points to many good ***authorities***



HITS example

- HITS continues until convergence
- Stop this example after 2 rounds



HITS vs. PageRank

- HITS is dependent on the query
  - You need the query to build the root set and base set
- PageRank is query-independent
  - One considers the links between pages, nothing else
  - Compute it once and save the result
- Scores from either/both algorithms can be combined with Vector model tf-idf similarity score

Agenda

- Document importance
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  - PageRank algorithm
  - HITS algorithm
- **Search engine optimization**

Search engine optimization

- *Search engine optimization:* techniques to increase the visibility of your page in search engine results

- Example 1: You have a personal web page. You want your web page to be the top hit when someone searches for your name.
- Example 2: Most users don't look beyond first page of search results. If your business appears on the first page, more clicks. More clicks -> more business.

Keywords

- TF x IDF is one factor in search rankings
- How can you influence TF x IDF?

Keywords

- TF x IDF is one factor in search rankings
- What can you do to influence TF x IDF?
  - Pages focus on a particular topic
  - You'll probably naturally use important (relevant) words multiple times
  - Length won't help you (remember normalization?)
  - Create a page with many repetitions of keywords
    - Search engines hate this. They "spam" filter for it.

Keyword technical considerations

- Search engine needs text to extract keywords
- Use text rather than images for important content
  - Is your name in a banner image? Is it also in the text?
- Site should work with JavaScript disabled
  - Some search engines use a JavaScript rendering engine on some pages, but don't count on it.

Links

- PageRank is another factor in search rankings
- What can you do to influence PageRank?

Link spam

- Bots add comments to blogs. Comments link back to your site.
- Blog owner should use `rel="nofollow"` attribute for links in comments. Search engines then ignore these links.
- Search engines detect and filter this as spam.

Link technical considerations

- The more legit links to your site the better
- Avoid links that look like form queries
  - http://www.mysite.com/info?about
- Use different links for different pages
  - Think about how wolverine access works. You click a link, and the URL doesn't change! Don't do this.

Other relevance factors

- Google says, “Relevancy is determined by over 200 factors, one of which is the PageRank for a given page.”

Next time

- Last time we used the words on a page to rank search results
  - Rank on document content
- Today we used the links between web pages to improve search results
  - Rank on document importance
- Next time we'll cover speed and scaling